

AWGN model: $x(t) = s_i(t) + w(t)$.

$P_e = \sum_{i=1}^M p_i P(e|m_i)$, P_e : total error rate, p_i : m_i 被选出的概率, $P(e|m_i)$: m_i 被判断错误的概率.

$$s_i(t) = \sum_{j=1}^N s_{ij} \phi_j(t).$$

$$s_i = [s_{i1}, s_{i2}, \dots, s_{iN}]^T, \quad \|s_i\| = \sqrt{\sum_{j=1}^N s_{ij}^2}, \quad E_i = \int_0^T s_i^2(t) dt = \sum_{j=1}^N s_{ij}^2$$

$$\int_0^T s_i(t) s_k(t) dt = s_i^T s_k \Rightarrow \text{时域内积} = \text{signal space 内积}.$$

$$d_{ik}^2 = \int_0^T (s_i(t) - s_k(t))^2 dt = \sum_{j=1}^N (s_{ij} - s_{kj})^2.$$

$$\cos \theta_{ik} = \frac{s_i^T s_k}{\|s_i\| \|s_k\|}.$$

$$\text{Gram-schmidt normalization: } \phi_1 = \frac{s_1(t)}{\sqrt{\int_0^T s_1^2(t) dt}}.$$

Orthonormal basis:

$$s_{12} = \int_0^T s_2(t) \phi_1(t) dt$$

$$g_1(t) = s_2(t) - s_{12} \phi_1(t).$$

$$\phi_2(t) = \frac{g_1(t)}{\sqrt{\int_0^T g_1^2(t) dt}}$$

BPSK:

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)$$

with $0 \leq t \leq T_b$.

$$s_2(t) = -\sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t)$$

E_b : bit energy, T_b : bit duration, f_c : carrier frequency.

$$f_c = \frac{n_c}{T_b}, \quad n_c \in \mathbb{N}.$$

$$\phi_1 = \sqrt{\frac{2}{T_b}} \cos(2\pi f_c t).$$

$$\Rightarrow \begin{cases} s_1 = \sqrt{E_b} \phi_1 \\ s_2 = -\sqrt{E_b} \phi_1 \end{cases}$$

receive signal: $x(t) = s_i(t) + w(t)$

$$x_1 = \int_0^{T_b} x(t) \cdot \phi_1(t) dt = s_{11} + n_1.$$

If 1 is sent:

$$x_1 = \sqrt{E_b} + n_1$$

If 0 is sent:

$$x_1 = -\sqrt{E_b} + n_1$$

$$n_1 \sim \mathcal{N}(0, \frac{N_0}{2}).$$



★ ML decision rule:

$$x_1 > 0 \Rightarrow 1$$

$$x_1 < 0 \Rightarrow 0$$

$$f_{x_1|m_i}(x_1|m_i) = \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{(x_1 - s_{11})^2}{N_0}\right), \quad \text{where } s_{11} = \sqrt{E_b}, \quad s_{12} = -\sqrt{E_b}.$$

$$P(1 \rightarrow 0) = P(x_1 < 0 | m_1) = P(n_1 < -\sqrt{E_b}) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

$$P(0 \rightarrow 1) = P(1 \rightarrow 0)$$

$$P_e = \frac{1}{2} P(1 \rightarrow 0) + \frac{1}{2} P(0 \rightarrow 1) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right).$$

QPSK:

$$s_i(t) = \sqrt{\frac{E}{T}} \cos \theta_i \cos(2\pi f_c t) - \sqrt{\frac{E}{T}} \sin \theta_i \sin(2\pi f_c t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_c t + \theta_i), \quad \theta_i \in \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}.$$

orthonormal basis function:

$$\phi_1(t) = \sqrt{\frac{T}{2}} \cos(2\pi f_c t)$$

$$\phi_2(t) = -\sqrt{\frac{T}{2}} \sin(2\pi f_c t)$$

$$s_i(t) = s_{i1} \phi_1(t) + s_{i2} \phi_2(t), \quad \text{where } s_{i1} = \sqrt{E} \cos \theta_i, \quad s_{i2} = \sqrt{E} \sin \theta_i.$$

Gray code and message point:

$$\theta_i \in \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}.$$

$$\text{分別對應 } \{10, 00, 01, 11\}.$$

$$(s_{i1}, s_{i2}) = (\sqrt{E} \cos \theta_i, \sqrt{E} \sin \theta_i):$$

$$10 \rightarrow (+, +)$$

$$00 \rightarrow (-, +)$$

$$01 \rightarrow (-, -)$$

$$11 \rightarrow (+, -)$$

Receive signal:

$$x(t) = s_i(t) + w(t).$$

$$x_1(t) = \int_0^T x(t) \cdot \phi_1(t) dt = s_{i1} + w_1$$

$$x_2(t) = \int_0^T x(t) \cdot \phi_2(t) dt = s_{i2} + w_2$$

Demodulation:

$$\star \text{ First bit: } x_1 > 0 \rightarrow 1, \quad x_1 < 0 \rightarrow 0$$

$$\text{Second bit: } x_2 > 0 \rightarrow 0, \quad x_2 < 0 \rightarrow 1$$

I 路、Q 路互相正交且不干擾，每一路可各自看成一個 BPSK。

$$\Rightarrow P_e' = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

$$\text{Total energy is } E_b, \text{ a symbol has two bits } \Rightarrow E_b = \frac{E}{2}.$$

$$\Rightarrow P_e' = Q\left(\sqrt{\frac{E}{N_0}}\right).$$

$$\text{The correct probability is } P_c = (1 - P_e')^2$$

$$\text{Then, symbol error rate is } P_s = 1 - P_c = 2P_e' - (P_e')^2.$$

$$\text{For Gray code QPSK, in high SNR, } P_b = \frac{P_s}{2} = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

OQPSK:

Q 路 delay $\frac{T}{2}$ symbol

$$\Rightarrow s_i(t) = s_I(t) \cos(2\pi f_c t) + s_Q(t - \frac{T}{2}) \sin(2\pi f_c t).$$

$$P_e^{\text{OQPSK}} = P_e^{\text{QPSK}}.$$

$\frac{\pi}{4}$ -shifted QPSK / DQPSK:

$$\theta_k = \theta_{k-1} + \Delta \theta_k, \quad \Delta \theta_k \in \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{4} \right\}.$$

$$x_1 = \int_0^T x(t) \phi_1(t) dt, \quad x_2 = \int_0^T x(t) \phi_2(t) dt$$

$$\Rightarrow \theta_k = \tan^{-1}\left(\frac{x_2}{x_1}\right)$$

利用 $\Delta \theta_k$ 來判斷 symbol.

$$\star \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, -\frac{3\pi}{4}, -\frac{\pi}{4} \right\} \text{ corresponding to } \{00, 01, 11, 10\}.$$

M-ary PSK: 訊號在圓用上

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos\left(2\pi f_c t + \frac{2\pi(i-1)}{M}\right), \quad i = 1, 2, \dots, M.$$

$$d_{\min} = 2\sqrt{E} \sin\left(\frac{\pi}{M}\right).$$

$$P_e \approx 2Q\left(\sqrt{\frac{2E}{N_0}} \sin\frac{\pi}{M}\right).$$

$$\star \text{ ML decision rule: } \hat{i} = \underset{i}{\operatorname{argmin}} \|x - s_i\|^2$$

QAM: 訊號在格點上

In (ϕ_1, ϕ_2) plane, $s_i = (a_i \frac{d_{min}}{2}, b_i \frac{d_{min}}{2})$

a_i, b_i are integers.

★ ML decision rule: $\hat{i} = \underset{i}{\operatorname{argmin}} \|x - s_i\|^2$

$$s_i(t) = a_i \frac{d_{min}}{2} \phi_1(t) + b_i \frac{d_{min}}{2} \phi_2(t).$$

square QAM:

$\log_2 M$ is even.

square QAM $\Rightarrow$ Cartesian product of two L-PAM ($L = \sqrt{M}$)

$$\Rightarrow a_i, b_i \in \{\pm 1, \pm 3, \dots, \pm(L-1)\}.$$

16 QAM:

$$M=16 \Rightarrow L=4.$$

$$a_i, b_i \in \{\pm 1, \pm 3\}.$$

$$\Rightarrow 16 \text{ QAM 的點: } (\pm 1, \pm 1), (\pm 1, \pm 3), (\pm 3, \pm 1), (\pm 3, \pm 3) \times \frac{d_{min}}{2}.$$

Using Gray coding for 16-QAM bit mapping.

SER:

$$\text{最小距離對列的能量 } E_0 \Rightarrow d_{min}^2 = 4E_0.$$

L-PAM 有內部點和外部點

$$P_{e,in} = 2 \cdot \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right), \quad P_{e,ex} = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right)$$

$$\Rightarrow P_e' = \frac{1}{L} ((L-2) P_{e,in} + 2 P_{e,ex})$$

$$= \frac{L-1}{L} \operatorname{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right).$$

MQAM: Two independent L-PAM

$$\Rightarrow P_c = (1 - P_e')^2$$

$$\Rightarrow P_e = 1 - P_c$$

$$\doteq 2 \left(1 - \frac{1}{\sqrt{M}}\right) \operatorname{erfc}\left(\sqrt{\frac{E_0}{N_0}}\right)$$

$$\text{Bandwidth efficiency: } \eta = \frac{R_b}{B} = \frac{\log_2 M}{2}$$

$$\text{Average symbol energy } E_{av}: E_{av} = \frac{4}{L} \sum_{i=1}^{\frac{L}{2}} (2i-1)^2 E_0 = \frac{1}{3} (L^2 - 1) E_0 = \frac{2}{3} (M-1) E_0$$

$$\Rightarrow E_0 = \frac{3 E_{av}}{2(M-1)}$$

$$\text{SER} \Rightarrow P_e \doteq 2 \left(1 - \frac{1}{\sqrt{M}}\right) \operatorname{erfc}\left(\sqrt{\frac{3 E_{av}}{2(M-1) N_0}}\right)$$

$$M=4 \Rightarrow \text{QPSK}.$$

For general pulse-shaped QAM and the k -th symbol.

$$c_k = a_k + jb_k.$$

Then, $s_k(t) = a_k g(t-kT) \cos(2\pi f_c t) - b_k g(t-kT) \sin(2\pi f_c t)$ and

$$s(t) = \sum_{k=-\infty}^{\infty} s_k(t).$$

$$= \operatorname{Re} \left\{ \sum_{k=-\infty}^{\infty} C_k g(t-kT) e^{j2\pi f_c t} \right\}.$$

($g(t)$ is baseband pulse, $C_k = a_k + jb_k$ is complex symbol)

$$\Rightarrow s(t) = \sum_{k=-\infty}^{\infty} a_k g_c(t-kT) - \sum_{k=-\infty}^{\infty} b_k \hat{g}_c(t-kT). \quad (\text{CAP form})$$

Matched filter:

$$\text{SNR}(T) = \frac{\left| \int_{-\infty}^{\infty} s(f) H(f) e^{j2\pi f T} df \right|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

$$\left| \int_{-\infty}^{\infty} s(f) H(f) e^{j2\pi f T} df \right|^2 \leq \int |s(f)|^2 df \cdot \int |H(f)|^2 df$$

$$" = " \text{ holds } \Rightarrow H_{\text{opt}}(f) = k S^*(f) e^{j2\pi f T}.$$

$$h_{\text{opt}}(t) = k s(T-t).$$

Coherent BFSK:

$$s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t), \quad 0 \leq t \leq T_b, \quad i=1,2.$$

$$f_i = \frac{n_i + \frac{1}{2}}{T_b}, \quad n_i \in \mathbb{N}, \quad i=1,2.$$

$$\phi_i(t) = \sqrt{\frac{2}{T_b}} \cos(2\pi f_i t), \quad 0 \leq t \leq T_b, \quad i=1,2.$$

$$s_{ij} = \int_0^{T_b} s_i(t) \phi_j(t) dt, \quad i,j=1,2.$$

$$\Rightarrow s_{11} = \sqrt{E_b}, \quad s_{12} = s_{21} = 0, \quad s_{22} = \sqrt{E_b}.$$

$$\Rightarrow s_1 = \begin{bmatrix} \sqrt{E_b} \\ 0 \end{bmatrix}, \quad s_2 = \begin{bmatrix} 0 \\ \sqrt{E_b} \end{bmatrix}.$$

$$d_{12} = \|s_1 - s_2\| = \sqrt{2E_b}.$$

Receive signal: $x(t) = s_i(t) + w(t)$.

$$x_1 = \int x(t) \phi_1(t) dt = s_{11} + w_1 \Rightarrow x = [x_1 \ x_2]^T$$

$$x_2 = \int x(t) \phi_2(t) dt = s_{12} + w_2$$

★ ML decision rule: $x_1 = x_2 \Rightarrow x_1 > x_2: s_1, \quad x_1 < x_2: s_2$.

$$\text{BER: } Y = x_1 - x_2$$

If 1 is sent:

$$E[Y|1] = E[x_1|1] - E[x_2|1] = \sqrt{E_b}.$$

If 0 is sent:

$$E[Y|0] = E[x_1|0] - E[x_2|0] = -\sqrt{E_b}.$$

$$\text{Var}[Y] = \text{Var}[x_1] + \text{Var}[x_2] = N_0.$$

PDF:

$$f_{Y|0}(y) = \frac{1}{\sqrt{2\pi N_0}} \exp\left[-\frac{(y - \sqrt{E_b})^2}{2N_0}\right].$$

$$f_{Y|1}(y) = \frac{1}{\sqrt{2\pi N_0}} \exp\left[-\frac{(y + \sqrt{E_b})^2}{2N_0}\right].$$

$$\Rightarrow P_e = Q\left(\sqrt{\frac{E_b}{N_0}}\right) = \frac{1}{2} \text{erfc}\left(\sqrt{\frac{E_b}{2N_0}}\right).$$

和 DPSK 比, 差 3 dB

$$\text{Central frequency: } f_c = \frac{f_1 + f_2}{2}.$$

$$f_1 = f_c + \frac{1}{2T_b}, \quad f_2 = f_c - \frac{1}{2T_b}.$$

$$\Rightarrow s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos\left(2\pi\left(f_c \pm \frac{1}{2T_b}\right)t\right).$$

$$= \sqrt{\frac{2E_b}{T_b}} \cos\left(\frac{\pi t}{T_b}\right) \cos(2\pi f_c t) \mp \sqrt{\frac{2E_b}{T_b}} \sin\left(\frac{\pi t}{T_b}\right) \sin(2\pi f_c t).$$

$$\text{Pulse shaping function: } g(t) = \sqrt{\frac{2E_b}{T_b}} \sin\left(\frac{\pi t}{T_b}\right)$$

$$G(f) = \sqrt{\frac{2E_b}{T_b}} \frac{1}{2j} \int_{-\infty}^{\infty} (e^{j\pi \frac{t}{T_b}} - e^{-j\pi \frac{t}{T_b}}) e^{-j\pi f t} dt$$

$$\Rightarrow \text{ESD: } |G(f)|^2 = \frac{8E_b T_b \omega^2 (\pi f T_b)^2}{\pi^2 (1 - 4f^2 T_b^2)^2} \sim \frac{1}{f^4}.$$

$$= \sqrt{\frac{2E_b}{T_b}} \frac{T_b (1 + e^{-j\pi f T_b})}{\pi(1 - 4f^2 T_b^2)} = \sqrt{\frac{2E_b}{T_b}} \frac{T_b \cdot 2e^{-j\pi f T_b} \cos(\pi f T_b)}{\pi(1 - 4f^2 T_b^2)}$$

CPFSK:

$$s(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \theta(t)), \quad \theta(t) = \theta(0) + \frac{\pi h}{T_b} t, \quad 0 \leq t \leq T_b.$$

$$f_1 = f_c + \frac{h}{2T_b}, \quad f_2 = f_c - \frac{h}{2T_b}.$$

$$\Rightarrow f_1 - f_2 = \frac{h}{T_b} \propto \frac{1}{T_b}, \quad h \text{ 用來控制 } f_1, f_2 \text{ 分多開.}$$

$$\Delta\theta = \theta(T_b) - \theta(0) = \begin{cases} \pi h, & 1 \text{ is sent} \\ -\pi h, & 0 \text{ is sent} \end{cases}$$

MSK:

CPFSK 的一種，條件： $h = \frac{1}{2}$ 。

$$\Rightarrow f_1 - f_2 = \frac{1}{2T_b}.$$

BER:

$$P_e^{\text{MSK}} = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

$$\Rightarrow P_e^{\text{MSK}} = P_e^{\text{BPSK}} = P_e^{\text{QPSK}}.$$

$$\text{PSD: } g(f) = \sqrt{\frac{2E_b}{T_b}} \cos\left(\frac{\pi f T_b}{2}\right).$$

$$\Rightarrow \text{ESD: } \eta_g = \frac{32E_b T_b \cos^2(\pi f T_b)}{\pi^2 (1 - 16f^2 T_b^2)^2} \sim \frac{1}{f^4}.$$

GMSK:

在 MSK 前加一個 Gaussian pulse shaping filter.

$$h(t) = \frac{\sqrt{\pi}}{\alpha} \exp\left(-\frac{\pi^2 t^2}{\alpha^2}\right), \quad \alpha = \sqrt{\frac{\ln 2}{2}} \cdot \frac{1}{W}.$$

$$H(f) = \exp\left(-\frac{\ln 2}{2} \left(\frac{f}{W}\right)^2\right).$$

$$\text{If } x(t) = u\left(t + \frac{T}{2}\right) - u\left(t - \frac{T}{2}\right), \quad g(t) = x(t) * h(t).$$

$$\Rightarrow g(t) = \frac{1}{2} \left[\operatorname{erfc}\left(\frac{\pi}{\alpha} \left(t + \frac{T}{2}\right)\right) - \operatorname{erfc}\left(\frac{\pi}{\alpha} \left(t - \frac{T}{2}\right)\right) \right]$$

BER:

$$P_e^{\text{GMSK}} = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$\text{SNR loss: } 10 \log \frac{\alpha}{2} \text{ dB.}$$

Coherent M-ary FSK:

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_i t), \quad i = 1, 2, \dots, M.$$

相鄰頻率只要相差 $\frac{1}{2T}$ 就可保持正交。

$$B = \frac{M}{2T}.$$

$$\text{Bandwidth efficiency: } \eta = \frac{2 \log_2 M}{M}.$$

★ ML decision rule: $\hat{i} = \operatorname{argmax}_i y_i$

$\Rightarrow$ 哪個 branch 最大，就判斷哪個 symbol.

Noncoherent BFSK:

$$s_i(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_i t), \quad i = 1, 2.$$

$$x(t) = \sqrt{\frac{2E}{T}} \cos(2\pi f_i t + \theta) + w(t)$$

$$\|x(t) - s_j\|^2 \triangleq \int_0^T \left(x(t) - \sqrt{\frac{2E}{T}} \cos(2\pi f_j t + \theta) \right)^2 dt.$$

$$f_X(x | s_j, \theta) = \frac{1}{\pi N_0} \exp\left(-\frac{1}{N_0} \|x - s_j\|^2\right)$$

$$s_1 = \begin{bmatrix} \sqrt{E_b} \\ 0 \end{bmatrix}, \quad s_2 = \begin{bmatrix} 0 \\ \sqrt{E_b} \end{bmatrix}.$$

$$x_j = \int_0^T x(t) \sqrt{\frac{2E}{T}} \cos(2\pi f_j t + \theta) dt = s_j + w_j, \quad j = 1, 2.$$

$$L(s_j; \theta) = \exp\left(\frac{\lambda}{N_0} \sqrt{\frac{2E}{T}} \int_0^T x(t) \cos(2\pi f_j t + \theta) dt\right).$$

$$\theta \text{ is a random variable: } f_\theta(\theta) = \begin{cases} \frac{1}{2\pi} & -\pi < \theta \leq \pi \\ 0 & \text{otherwise.} \end{cases}$$

$$L(s_i) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp\left(\frac{\lambda}{N_0} \sqrt{\frac{2E}{T}} \int_0^T x(t) \cos(2\pi f_i t + \theta) dt\right) d\theta \Rightarrow \text{Not dependent on } \theta.$$

$$x_{Ii} = \int_0^T x(\tau) \cos(2\pi f_i \tau) d\tau$$

$$x_{Qi} = \int_0^T x(\tau) \sin(2\pi f_i \tau) d\tau$$

$$\text{Let } \lambda_i = \sqrt{x_{Ii}^2 + x_{Qi}^2}, \quad \beta_i = \tan^{-1} \left(\frac{x_{Qi}}{x_{Ii}} \right)$$

$$\Rightarrow x_{Ii} = \lambda_i \cos \beta_i, \quad x_{Qi} = \lambda_i \sin \beta_i$$

$$\begin{aligned} \int_0^T x(\tau) \cos(2\pi f_i \tau + \theta) d\tau &= \lambda_i \cos \beta_i \cos \theta - \lambda_i \sin \beta_i \sin \theta \\ &= \lambda_i \cos(\beta_i + \theta) \end{aligned}$$

$$\text{Then, } L(s_i) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \exp \left(\frac{j}{N_0} \sqrt{\frac{E}{T}} \lambda_i \cos \theta \right) d\theta = I_0 \left(\frac{j}{N_0} \sqrt{\frac{E}{T}} \lambda_i \right)$$

$\Rightarrow L(s_i)$ is the modified Bessel function of the first kind of order zero.

★ ML decision rules:

$$L(s_2) > L(s_1) : m_2, \quad L(s_1) < L(s_1) : m_1$$

I_0 is monotonic increasing for the argument

$$\Rightarrow \lambda_2 > \lambda_1 : m_2, \quad \lambda_2 < \lambda_1 : m_1$$

Matched filter:

If filter matched to $s(\tau) = \cos(2\pi f_i \tau + \theta)$,

the the output is

$$\begin{aligned} y(t) &= \int_0^T x(\tau) \cdot \cos(2\pi f_i (T-t+\tau) + \theta) d\tau \\ &= \cos(2\pi f_i (T-t) + \theta) \int_0^T x(\tau) \cos(2\pi f_i \tau) d\tau \\ &\quad - \sin(2\pi f_i (T-t) + \theta) \int_0^T x(\tau) \sin(2\pi f_i \tau) d\tau \end{aligned}$$

$$\text{When } t = T, \quad y(T) = \lambda_i \cos(\beta_i + \theta)$$

$\Rightarrow$ Envelope doesn't depend on the phase θ .

This prove that quadrature receiver is equivalent to matched filter + envelope detector.

BER:

$$\text{sent } s_1(t) \rightarrow \lambda_2 > \lambda_1 \rightarrow \text{error}$$

Since $s_2(t)$ is orthogonal to $s_1(t)$,

$$x_{I2} \sim N(0, \frac{N_0}{2}), \quad x_{Q2} \sim N(0, \frac{N_0}{2})$$

$$\Rightarrow f_{x_{I2}, x_{Q2}}(x_{I2}, x_{Q2}) = \frac{1}{\pi N_0} \exp \left(-\frac{x_{I2}^2 + x_{Q2}^2}{N_0} \right)$$

$$\Rightarrow f_{L_2}(\lambda_2) = \begin{cases} \frac{\lambda_2}{N} \exp \left(-\frac{\lambda_2^2}{N_0} \right), & \lambda_2 > 0 \\ 0, & \text{otherwise} \end{cases} \Rightarrow \text{Rayleigh distribution.}$$

$$\Rightarrow P(\lambda_2 > \lambda_1 | \lambda_1) = \int_{\lambda_1}^{\infty} f_{L_2}(\lambda_2) d\lambda_2 = \exp \left(-\frac{\lambda_1^2}{N_0} \right)$$

$$x_{I1} \sim N(\sqrt{E}, \frac{N_0}{2}), \quad x_{Q1} \sim N(0, \frac{N_0}{2})$$

$$\Rightarrow f_{x_{I1}, x_{Q1}}(x_{I1}, x_{Q1}) = \frac{1}{\pi N_0} \exp \left(-\frac{(x_{I1} - \sqrt{E})^2 + x_{Q1}^2}{N_0} \right)$$

$$P_e = \iint P(\text{error} | x_{I1}, x_{Q1}) \cdot f_{x_{I1}, x_{Q1}}(x_{I1}, x_{Q1}) dx_{I1} dx_{Q1}$$

$$= \frac{1}{2} \exp \left(-\frac{E_b}{2N_0} \right)$$

BER Table :

Coherent BPSK / QPSK / MSK : $\frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$

Coherent BFSK : $\frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$

DPSK : $\frac{1}{2} \exp\left(-\frac{E_b}{N_0}\right)$

Noncoherent BFSK : $\frac{1}{2} \exp\left(-\frac{E_b}{2N_0}\right)$

4.

$$s_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t)$$

$$x_i(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_i t + \theta) + w(t)$$

$$x_{2i} = \int_0^{T_b} x_i(t) \cos(2\pi f_i t) dt$$

$$x_{0i} = \int_0^{T_b} x_i(t) \sin(2\pi f_i t) dt$$

$$l_i^2 = x_{2i}^2 + x_{0i}^2, \quad \beta_i = \tan^{-1}\left(\frac{x_{0i}}{x_{2i}}\right)$$

$$\Rightarrow l_1 > l_2 : m_1, \quad l_2 > l_1 : m_2$$

$$P_e = P(l_2 > l_1 | s_1)$$

$$x_{21} \sim N(0, \frac{N_0}{2}), \quad x_{01} \sim N(0, \frac{N_0}{2}), \quad \Rightarrow (x_{21}, x_{01}) \sim N(0, \frac{N_0}{2})$$

$$f_{x_{21}, x_{01}}(x_{21}, x_{01}) = \frac{1}{\pi N_0} \exp\left(-\frac{x_{21}^2 + x_{01}^2}{N_0}\right) \quad l_2 \text{ is Rayleigh}$$

$$f_{l_2, \beta_2}(l_2, \beta_2) = \frac{l_2}{\pi N_0} \exp\left(-\frac{l_2^2}{N_0}\right) \quad P(l_2 > l_1 | l_1) = \exp\left(-\frac{l_1^2}{N_0}\right)$$

$$f_{l_1} = \int_{-\pi}^{\pi} \frac{1}{\pi N_0} \exp\left(-\frac{l_1^2}{N_0}\right) d\beta_1 = \frac{2l_1}{N_0} \exp\left(-\frac{l_1^2}{N_0}\right)$$

$$P(l_2 > l_1 | l_1) = \int_{l_1}^{\infty} f_{l_2} dl_2 = \exp\left(-\frac{l_1^2}{N_0}\right)$$

$$x_{21} \sim N(\sqrt{E_b}, \frac{N_0}{2}), \quad x_{01} \sim N(0, \frac{N_0}{2})$$

$$f_{x_{21}, x_{01}}(x_{21}, x_{01}) = \frac{1}{\pi N_0} \exp\left(-\frac{(x_{21} - \sqrt{E_b})^2 + x_{01}^2}{N_0}\right)$$

$$P(\text{error} | x_{21}, x_{01}) = \exp\left(-\frac{x_{21}^2 + x_{01}^2}{N_0}\right)$$

$$x_{21} \sim N(\sqrt{E_b}, \frac{N_0}{2}), \quad x_{01} \sim N(0, \frac{N_0}{2})$$

$$\Rightarrow f_{x_{21}, x_{01}}(x_{21}, x_{01}) = \frac{1}{\pi N_0} \exp\left(-\frac{(x_{21} - \sqrt{E_b})^2 + x_{01}^2}{N_0}\right)$$

$$P_e = \iint P(\text{error} | x_{21}, x_{01}) f_{x_{21}, x_{01}}(x_{21}, x_{01}) dx_{21} dx_{01}$$

$$= \frac{1}{\pi N_0} \iint \exp\left(-\frac{x_{21}^2 + x_{01}^2}{N_0}\right) \exp\left(-\frac{(x_{21} - \sqrt{E_b})^2 + x_{01}^2}{N_0}\right) dx_{21} dx_{01}$$

$$= \frac{1}{\pi N_0} e^{-\frac{E_b}{2N_0}} \int \exp\left(-\frac{1}{N_0}(x_{21} - \frac{\sqrt{E_b}}{2})^2\right) dx_{21} \int \exp\left(-\frac{2x_{01}^2}{N_0}\right) dx_{01}$$

$$= \frac{1}{2} e^{-\frac{E_b}{2N_0}}$$

1-1

$$w_1 \sim N(0, \frac{N_0}{2})$$

$$x_1 | s_1 \sim N(\sqrt{E_b}, \frac{N_0}{2})$$

$$x_1 | s_2 \sim N(-\sqrt{E_b}, \frac{N_0}{2})$$

$$f_{x_1}(x_1 | s_1) = \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{(x_1 - \sqrt{E_b})^2}{N_0}\right)$$

$$f_{x_1}(x_1 | s_2) = \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{(x_1 + \sqrt{E_b})^2}{N_0}\right)$$

$$P(s_2 | s_1) = P(x_1 < 0 | s_1)$$

$$= P(\sqrt{E_b} + w_1 < 0)$$

$$= P(w_1 < -\sqrt{E_b})$$

$$= P(z < -\sqrt{\frac{E_b}{N_0}})$$

$$= Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$= \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$z = \frac{w_1}{\sqrt{N_0/2}} \sim N(0, 1)$$

$$\text{Similarly, } P(s_1 | s_2) = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$P_e = \frac{1}{2} P(s_2 | s_1) + \frac{1}{2} P(s_1 | s_2) = \frac{1}{2} \operatorname{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right)$$

1-2

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t)$$

$$\phi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t)$$

$$s_i(t) = s_{i1} \phi_1(t) + s_{i2} \phi_2(t)$$

$$s_{i1} = \pm \sqrt{E_b}, \quad s_{i2} = \pm \sqrt{E_b}$$

$$\text{Since a symbol has 2 bits, } E_b = \frac{E}{2}$$

$$x(t) = s_i(t) + w(t)$$

$$x_1 = \int x(t) \phi_1(t) dt = s_{i1} + w_1$$

$$x_2 = \int x(t) \phi_2(t) dt = s_{i2} + w_2$$

$$w_1 \sim N(0, \frac{N_0}{2}), \quad w_2 \sim N(0, \frac{N_0}{2})$$

$$\text{Since each branch of QPSK is like BPSK and bit energy } E_b = \frac{E}{2}$$

$$P_b' = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$P_e = (1 - P_b')^2$$

$$P_s = 1 - P_e = 2P_b' - (P_b')^2$$

$$\text{In high SNR, } P_s \approx 2P_b'$$

$$\text{Thus, } P_s \approx 2Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$\text{Since for Gray-code QPSK, the adjacent points only differ by a bit}$$

$$\text{Thus, } P_e = \frac{1}{2} P_s \approx Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

1-3

$$s(t) = s_1 \phi_1(t) + s_2 \phi_2(t)$$

$$x_1 = s_1 + n_1, \quad x_2 = s_2 + n_2$$

$$n_1 \sim \mathcal{N}(0, \frac{N_0}{2}), \quad n_2 \sim \mathcal{N}(0, \frac{N_0}{2}).$$

Each branch is equivalent to coherent BPSK

$$\Rightarrow P_b' = P_b^{\text{BPSK}}$$

$$P_c = (1 - P_b')^2$$

$$P_s = 1 - P_c \approx 2P_b'$$

4 messages are equally likely, as long as one branch is wrong, the bit may be decided to the other one, and a branch has a symbol, i.e. two bits.

$$\Rightarrow P_e = \frac{1}{4} \cdot 2 P_s = P_b' = Q\left(\sqrt{\frac{2E_b}{N_0}}\right).$$

2.

$$s_1(t) = \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_1 t), \quad s_2(t) = \sqrt{\frac{2E_b}{T}} \cos(2\pi f_2 t).$$

$$\Rightarrow \text{signal point } (\sqrt{E_b}, 0), (0, \sqrt{E_b})$$

$$\text{received: } x_1, x_2$$

$$\text{ML decision rule: } x_1 = x_2 \Rightarrow x_1 > x_2 : s_1, \quad x_1 < x_2 : s_2.$$

$$\text{BER: } Y = x_1 - x_2$$

If 1 is sent:

$$x_1 = \sqrt{E_b} + w_1, \quad x_2 = w_2$$

$$E[Y|1] = E[x_1|1] - E[x_2|1] = \sqrt{E_b}.$$

If 0 is sent:

$$E[Y|0] = E[x_1|0] - E[x_2|0] = -\sqrt{E_b}.$$

$$\text{Var}[Y] = \text{Var}[x_1] + \text{Var}[x_2] = N_0.$$

$$\Rightarrow Y|s_1 \sim \mathcal{N}(\sqrt{E_b}, N_0)$$

$$Y|s_2 \sim \mathcal{N}(-\sqrt{E_b}, N_0)$$

$$P(s_2|s_1) = P(Y < 0|s_1)$$

$$= P\left(Z < -\frac{\sqrt{E_b}}{\sqrt{N_0}}\right)$$

$$= Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$= \frac{1}{2} \text{erfc}\left(\sqrt{\frac{E_b}{2N_0}}\right)$$

$$Z = \frac{Y - \sqrt{E_b}}{\sqrt{N_0}} \sim \mathcal{N}(0, 1)$$

$$\text{Similarly, } P(s_1|s_2) = \frac{1}{2} \text{erfc}\left(\sqrt{\frac{E_b}{2N_0}}\right)$$

$$P_e = \frac{1}{2} (P(s_1|s_2) + P(s_2|s_1)) = \frac{1}{2} \text{erfc}\left(\sqrt{\frac{E_b}{2N_0}}\right)$$

3.

$$\text{Sent 1: } S_1(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & , 0 \leq t \leq T_b \\ \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & , T_b \leq t \leq 2T_b \end{cases}$$

$$\text{Sent 0: } S_2(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t) & , 0 \leq t \leq T_b \\ \sqrt{\frac{2E_b}{T_b}} \cos(2\pi f_c t + \pi) & , T_b \leq t \leq 2T_b \end{cases}$$

$$x(t) = \pm A \cos(2\pi f_c t + \theta) + w(t).$$

$$x_I = \frac{2}{T_b} \int_0^{T_b} x(t) \cos(2\pi f_c t) dt$$

$$x_Q = -\frac{2}{T_b} \int_0^{T_b} x(t) \sin(2\pi f_c t) dt$$

$$\Rightarrow \begin{cases} x_I = \pm A \cos \theta + n_I \\ x_Q = \pm A \sin \theta + n_Q \end{cases}$$

$$x_0 = \begin{bmatrix} x_{I0} \\ x_{Q0} \end{bmatrix}, \quad x_1 = \begin{bmatrix} x_{I1} \\ x_{Q1} \end{bmatrix}$$

$$\text{If symbol 1 is sent: } x_0^T x_1 \neq A^2$$

$$\text{If symbol 0 is sent: } x_0^T x_1 \neq -A^2$$

$$\Rightarrow \text{decision rule: } x_0^T x_1 > 0 : 1, \quad x_0^T x_1 < 0 : 0$$

$$l_1^2 = (\overset{L_{I1}}{x_{I0} + x_{I1}})^2 + (\overset{L_{Q1}}{x_{Q0} + x_{Q1}})^2$$

$$l_2^2 = (\overset{L_{I2}}{x_{I1} - x_{I1}})^2 + (\overset{L_{Q2}}{x_{Q0} - x_{Q1}})^2$$

$$4 x_1^T x_0 = l_1^2 - l_2^2$$

$$\Rightarrow \text{decision rule becomes } l_1^2 > l_2^2 : 1, \quad l_1^2 < l_2^2 : 0.$$

When 1 is sent:

$$L_{I1} \sim N(\pm \sqrt{E_b} \cos \theta, N_0) = N(\mu_I, N_0)$$

$$L_{Q1} \sim N(\pm \sqrt{E_b} \sin \theta, N_0) = N(\mu_Q, N_0)$$

$$L_{I2} \sim N(0, N_0)$$

$$L_{Q2} \sim N(0, N_0)$$

$$P_e' = P(l_2 > l_1 | l_1)$$

$$l_2 = \sqrt{L_{I2}^2 + L_{Q2}^2}$$

$$(L_{I2}, L_{Q2}) \text{ is } \sim N(0, N_0)$$

$$\Rightarrow l_2 \text{ is Rayleigh form}$$

$$\Rightarrow P_e' = \exp\left(-\frac{l_1^2}{4N_0}\right)$$

$$(L_{I1}, L_{Q1}) \text{ is } \sim N(4E_b, N_0)$$

$$f_{L_{I1}, L_{Q1}}(l_{I1}, l_{Q1}) = \frac{1}{2\pi N_0} \exp\left(-\frac{(l_{I1} - \mu_I)^2 + (l_{Q1} - \mu_Q)^2}{2N_0}\right)$$

$$P_e = \iint P_e' f_{L_{I1}, L_{Q1}}(l_{I1}, l_{Q1}) dl_{I1} dl_{Q1}$$

$$= \frac{1}{2\pi N_0} \iint \exp\left(-\frac{l_1^2 + (l_{I1} - \mu_I)^2 + (l_{Q1} - \mu_Q)^2}{2N_0}\right) dl_{I1} dl_{Q1}$$

$$\begin{aligned}
 & \lambda_1^2 + (\lambda_{T1} - \mu_T)^2 + (\lambda_{Q1} - \mu_Q)^2 \\
 &= 2\lambda_{T1}^2 - 2\mu_T \lambda_{T1} + \mu_T^2 + 2\lambda_{Q1}^2 - 2\mu_Q \lambda_{Q1} + \mu_Q^2 \\
 &= 2\left(\lambda_{T1} - \frac{\mu_T}{2}\right)^2 + \frac{\mu_T^2}{2} + 2\left(\lambda_{Q1} - \frac{\mu_Q}{2}\right)^2 + \frac{\mu_Q^2}{2}
 \end{aligned}$$

$$\Rightarrow p_e = \frac{1}{2\pi N_0} \frac{\exp\left(-\frac{\mu_T^2 + \mu_Q^2}{4N_0}\right)}{\exp\left(-\frac{E_b}{N_0}\right)} \frac{\iint \exp\left(-\frac{(\cdot)^2 + (\cdot)^2}{N_0}\right) d\cdot d\cdot}{\sqrt{\pi N_0} \cdot \sqrt{\pi N_0}}$$

$$\int_{\mathbb{R}} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \quad .$$

$$= \frac{1}{2} \exp\left(-\frac{E_b}{N_0}\right) \quad .$$